

Challenges of the Generative AI

1. **Toxicity** is the possibility of generating content (whether it be text, images, or other modalities) that is offensive, disturbing, or otherwise inappropriate. This is a primary concern with generative AI.

It is hard to even define and scope the toxicity.

2. **Hallucinations** are assertions or claims that sound plausible but are verifiably incorrect. Considering the next-word distribution sampling employed by large language models (LLMs), it is perhaps not surprising that in more objective or factual use cases, LLMs are susceptible to hallucinations.

For example, a common phenomenon with current LLMs is creating nonexistent scientific citations. Suppose that an LLMs is prompted with the request, "Tell me about some papers by" a particular author. The model is not actually searching for legitimate citations but generating ones from the distribution of words associated with that author. The result might include realistic titles and topics in the area of the author. However, these might not be real articles, and they might include plausible coauthors but not actual ones.

3. **Intellectual property (IP)**: focuses on who own content generated by AI, whether AI-driven inventions are patentable, and the legality of using the copyrighted data to train models. Generally, current laws require human authorship, meaning purely AI-generated outputs often lack protection.
4. **Plagiarism and cheating**: The creative capabilities of generative AI give rise to worries that it will be used to write college essays, writing samples for job applications, and other forms of cheating or illicit copying. Debates on this topic are happening at universities and many other institutions, and attitudes vary widely.
5. **Disruption of the nature of work**: The proficiency with which generative AI is able to create compelling text and images, perform well on standardized tests, write entire articles on given topics, and successfully summarize or improve the grammar of provided articles has created some anxiety. There is a concern that some professions might be replaced or seriously disrupted by the technology.